

Cumulative sit-out penalty

Pickleball.ai scheduler — analysis of how session history affects who gets benched. Prepared 15 August 2026.

The question, answered

The penalty compares **raw counts**, not rates. A player with no history defaults to **zero**.

This is the whole calculation, from the scoring function:

```
const lifetimeSitOuts =
  historyContext.pastSitOuts[index]
  + schedule.sitOutTally[index];
```

A lifetime tally added to tonight’s tally. Nothing is divided by sessions attended. The penalty is then the spread — highest total minus lowest — multiplied by 6.

The zero default comes from how the tally is created: an array of one slot per player, filled with 0. Anyone absent from the record is simply never incremented.

Why the combination is wrong

Raw counts reward attendance rather than measuring fairness, because somebody who comes rarely cannot accumulate sit-outs. Two findings, both measured over 20 generated schedules with one bench place per round.

Finding 1 — newcomers are penalised

	Past sit-outs	Share of bench time	Fair share
Four regulars	48 each	75% combined	80%
Newcomer (first night)	0	25%	20%

A zero record reads as “maximally owed bench time”, so the newcomer is benched more than her fair share — penalised precisely for having never played.

Finding 2 — fairness is inverted

	Sessions	Raw sit-outs	Per session	Benched (20 runs)
Faithful	8	8	1.0	17
Occasional	2	6	3.0	20

Occasional is treated three times worse every night she attends, yet has *fewer* raw sit-outs than Faithful — so the scheduler benches her even more. This is the more serious of the two problems.

The flaw only bites with irregular attendance. In a stable group where everyone turns up most weeks, raw counts and rates agree closely.

Three ways forward

“Fair” here is a judgment about how the club runs, not a technical detail — hence a decision rather than a silent fix.

- **Option A — compare rates.** Use sit-outs divided by sessions attended, so Occasional's 3.0 correctly outranks Faithful's 1.0. Newcomers need a starting value; the section average is the neutral choice, meaning “no claim either way” rather than “maximally owed”. Fixes both findings.
- **Option B — keep raw counts, seed newcomers at the average.** A smaller change that fixes Finding 1, but leaves the Faithful/Occasional inversion in Finding 2 intact.
- **Option C — leave it.** Defensible if attendance is stable. No complexity added for a case that does not arise in practice.

Nothing has been changed in the scheduler. The behaviour described here is what is currently committed and running.